Sols

Math 161 Calculus I — Practice for Exam #2

True or False. ~~No work/explanation required~~. **True means always true.**

1. *If a function is continuous, it is always differentiable.*

False, $y = |x|$ is an example of a continuous function which is not differentiable at $x=0$.

2. *A critical point c is only where $f'(c) = 0$.*

False, also where $f'(c)$ is undefined

3. *If $f'(c) = 0$, then f has a local maximum or minimum at c.*

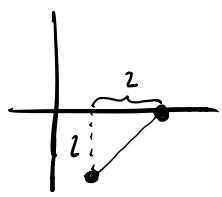
False, Could be an inflection point or asymptote

4. *If $f''(c) = 0$, then $x = c$ is an inflection point of f.*

False, must check concavity switches

5. *There exists a function f such that $f(1) = -2$, $f(3) = 0$, and $f'(x) > 1$ for all x.*

False,



At best the function is a line with $m=1$.

6. *Find $h'(2)$, given that $f(2) = -3$, $g(2) = 4$, $f'(2) = -2$, and $g'(2) = 7$, if $h(x) = \dfrac{g(x)}{1 + f(x)}$.*

$$h'(x) = \frac{g'(x)(1+f(x)) - g(x)f'(x)}{(1+f(x))^2}$$

$$h'(2) = \frac{7(1-3) - 4(-2)}{(-2)^2} = \frac{-14+8}{4} = \frac{-6}{4} = -3/2$$

7. *A student turns in the incorrect solution to the problem below. Explain the student's mistake in words, using complete sentences. Then work out the correct solution.*

$$\frac{d}{dx}(x^2 \tan y) = 2x \tan y$$

The student did not use the product rule and also forgot to use implicit differentiation.

$$2x \tan y + x^2 \sec^2 y \frac{dy}{dx}$$

8. *Find the value or values of c that satisfy the equation*

$$\frac{f(b)-f(a)}{b-a} = f'(c)$$

in the conclusion of the Mean Value Theorem for the function $f(x) = x^2 + 2x - 1$ *on* $[0, 1]$.

$$\frac{f(1)-f(0)}{1-0} = \frac{2+1}{1} = 3$$

$$f'(c) = 2c + 2$$

$$2c+2 = 3$$

$$2c = 1$$

$$\boxed{c = 1/2}$$

9. *When a circular plate of metal is heated in an oven, its radius increases at a rate of 0.01 cm/min. At what rate is the plate's area increasing when the radius is 50cm? (Recall the area of a circle is given by* $A = \pi r^2$.*)*

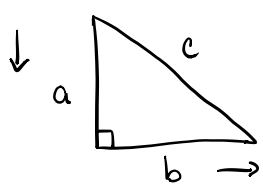
$$A = \pi r^2$$

$$\frac{dA}{dt} = 2\pi r \frac{dr}{dt}$$

$$\frac{dA}{dt} = 2\pi(50)(.01)$$

$$\frac{dA}{dt} = \pi \ \text{cm}^2/\text{min}$$

10. *The top of a ladder slides down a vertical wall at a rate of* $0.15m/s$. *At the moment when the bottom of the ladder is* $3m$ *from the wall, it slides away from the wall at a rate of* $0.2m/s$. *How long is the ladder?*



$$a^2 + b^2 = c^2$$

$$2a\frac{da}{dt} + 2b\frac{db}{dt} = 2c\frac{dc}{dt} = 0$$

$$2a(-.15) + 2(3)(.2) = 0$$

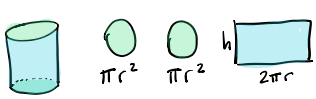
$$-.3a = -1.2 \qquad a = 4m$$

$$4^2 + 3^2 = c^2$$

$$\boxed{c = 5m}$$

Optimization

11. *A cylindrical can is to be made to hold 1 L of oil. Find the dimensions that will minimize the cost of the metal to manufacture the can.*



$$SA = 2\pi r^2 + 2\pi r h$$

$$SA = 2\pi r^2 + 2\pi r\left(\frac{1000}{\pi r^2}\right)$$

$$= 2\pi r^2 + \frac{2000}{r}$$

$$1L = 1000\ \text{cm}^3$$

$$\pi r^2 h = 1000$$

$$h = \frac{1000}{\pi r^2}$$

$$(SA)' = 4\pi r - \frac{2000}{r^2}$$

$$= \frac{4\pi r^3 - 2000}{r^2}$$

$(SA)' = 0$ when $r = \sqrt[3]{500/\pi}$

undefined when $r = 0$

$$\boxed{r = \sqrt[3]{\frac{500}{\pi}}\ \text{cm}}$$

12. *Sketch the curve*

$$y = \frac{x^2 - 4}{2x}$$

(a) State the domain.

$(-\infty, 0) \cup (0, \infty)$

(b) Find the intercepts. Enter NONE if there are none.

x-intercepts: $x = 2, -2$ $(2,0), (-2,0)$

y-intercept: None

(c) Is the function even, odd, or neither? What type of symmetry does the function have?

$f(-x) = \frac{x^2-4}{-2x} = -\frac{x^2-4}{2x} = -f(x)$ Odd

Symmetric about origin

(d) Find the asymptotes. Enter NONE if there are none.

Horizontal: None

Oblique: $y = \frac{1}{2}x$

Vertical: $x = 0$

Must show this work!

$\lim_{x \to 0^-} \frac{x^2-4}{2x} = \infty$

$\lim_{x \to 0^+} \frac{x^2-4}{2x} = -\infty$

$x = 0$

$$\begin{array}{r} \frac{1}{2}x \\ 2x \overline{)\, x^2 - 4} \\ x^2 \\ \hline -4 \end{array}$$

(e) Find the intervals where the function is increasing and decreasing. Enter NONE if not applicable.

Increasing: $(-\infty, 0) \cup (0, \infty)$

Decreasing: NONE

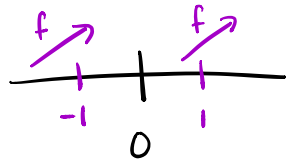
$$f'(x) = \frac{2x(2x) - (x^2-4)2}{4x^2}$$

$$= \frac{4x^2 - 2x^2 + 8}{4x^2}$$

$$= \frac{2x^2+8}{4x^2} = \frac{x^2+4}{2x^2}$$

$f'(x)$ is never 0

$f'(x)$ is undefined when $x=0$



$f'(-1) > 0$

$f'(1) > 0$

(f) State the local maximum and local minimum value(s). Enter NONE if not applicable.

Local maximum value(s): NONE

Local minimum value(s): NONE

(g) Find the intervals on which the function is concave up and concave down. State the inflection points. Enter NONE if not applicable.

Concave Up: $(-\infty, 0)$

Concave Down: $(0, \infty)$

Inflection Points: NONE ($x=0$ is not in the domain)

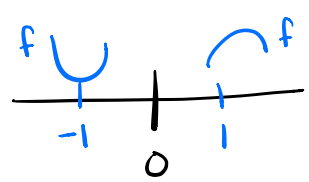
$$f''(x) = \frac{2x(2x^2) - (x^2+4)(4x)}{4x^4}$$

$$= \frac{4x^3 - 4x^3 - 16x}{4x^4}$$

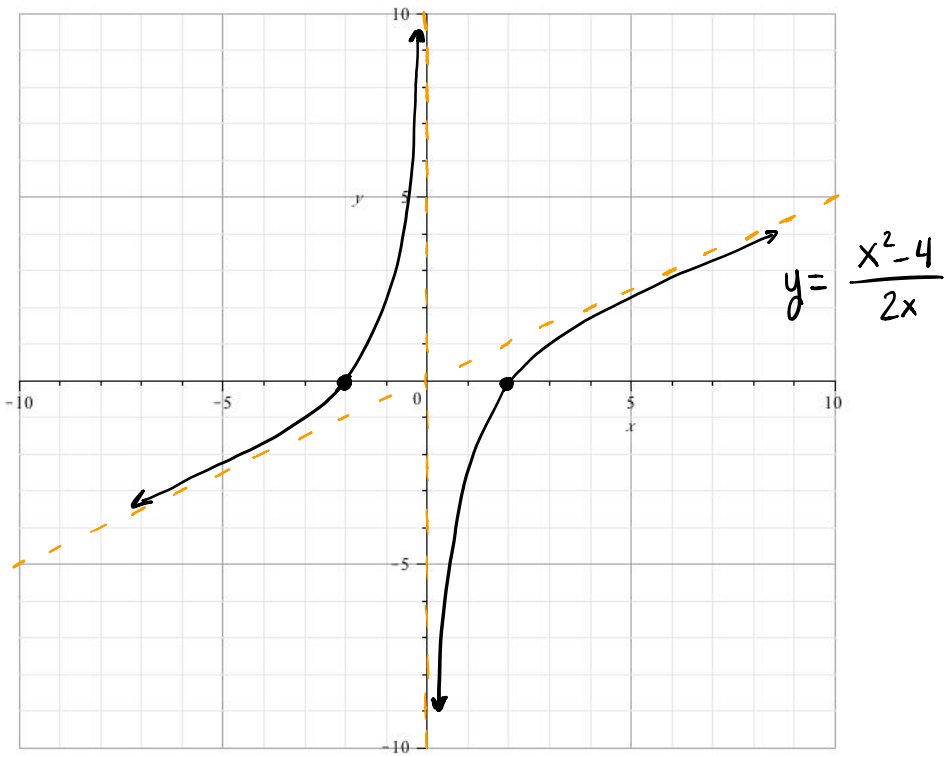
$$= \frac{-16x}{4x^4} = \frac{-4}{x^3}$$

$f''(x)$ is never 0

$f''(x)$ undefined when $x=0$



$f''(-1) > 0$

$f''(1) < 0$

(h) Use parts (a)-(g) to sketch the curve. Be sure that your graph is labeled and neat. Messy/incoherent graphs will receive zero points.